DRUGS · NASAL PREPARATIONS

Steroid drops and irrigations

Sprays reach the front of the nose and the inferior turbinate but deliver little to the middle meatus, olfactory cleft and sinuses, where polyps and chronic rhinosinusitis live. Steroid drops given in a head-down position, fluticasone nasules and budesonide added to high-volume saline irrigation deliver more steroid to those sites. They are used for nasal polyps and chronic rhinosinusitis, especially after sinus surgery, and in loss of smell.

1 CHAPTER ONE At a glance

The treatments on one page.

1.1 Summary

STEROID DROPS AND IRRIGATIONS

	SUMMARY
Preparations	Betamethasone sodium phosphate 0.1% drops; fluticasone propionate nasules (400 micrograms per unit dose); budesonide respules (0.5 to 1 mg) added to a saline rinse bottle (off-label); steroid-releasing sinus implants (mometasone)
ENT uses	Nasal polyps; chronic rhinosinusitis after surgery; olfactory loss from polyps or eosinophilic disease; postoperative oedema
Doses	Betamethasone 2 to 3 drops into each nostril two or three times daily in the head-down position, for short courses (commonly up to 4 to 6 weeks); fluticasone nasules one 400 microgram unit once or twice daily, divided between the nostrils; budesonide 0.5 to 1 mg in 240 ml saline once or twice daily
Avoid	Prolonged betamethasone drops (systemic absorption and adrenal suppression)

2 CHAPTER TWO Using them in ENT

Technique and evidence.

2.1 Positions for drops

Drops must run along the roof of the nose: the **head-down and forward** ("praying to Mecca") position, the **lying head back over the edge of the bed** (Mygind) position, or lying on one side with the head turned down (lateral head low). Hold the position for 1 to 2 minutes for each nostril. The upright position wastes the dose down the throat.

2.2 Choosing the agent

Betamethasone drops are potent and work well for short courses, but are absorbed systemically; repeated courses can suppress the adrenal axis and cause Cushingoid features, so limit their use and reserve them for flares or before surgery. **Fluticasone nasules** have low bioavailability and suit longer use. **Budesonide rinses** after sinus surgery deliver steroid widely with acceptable systemic absorption and improve symptom scores and endoscopic appearance.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- With betamethasone drops: **adrenal suppression**, raised blood glucose, glaucoma and cataract risk with prolonged use.
- Local irritation, epistaxis.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Why drops and rinses reach the sinuses better than sprays.
- The head-down positions.
- The risk of systemic absorption with betamethasone drops.

4.2 Common pitfalls

COMMON PITFALLS

- Prescribing betamethasone drops indefinitely.
- Instilling drops with the head upright.

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